

HAT-P-18b

R-2229

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-18_210507 · created 2026-08-15T00:59:35+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459341.76768 +/- 0.00027 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.122 +/- 0.012
Transit depth (Rp/Rs)^2	1.48 +/- 0.29 [%]
Orbital Inclination (inc)	89.96 +/- 0.48
Airmass coefficient 1 (a1)	0.96 +/- 0.014
Airmass coefficient 2 (a2)	0.0296 +/- 0.0043
Scatter in the residuals of the lightcurve fit is	1.51 %
Best Comparison Star	#2 - [254, 333]
Optimal Aperture	3.72
Optimal Annulus	9.98
Transit Duration (day)	0.1166 +/- 0.0018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.122 ± 0.012 vs 0.1356 (deviation 1.11σ)
Duration fit vs published	0.1166 d vs 0.1129 d (deviation 2.0σ)
Mid-time O–C	0.2 min
Depth SNR	9.9
Residual scatter	1.51 %

Light curve

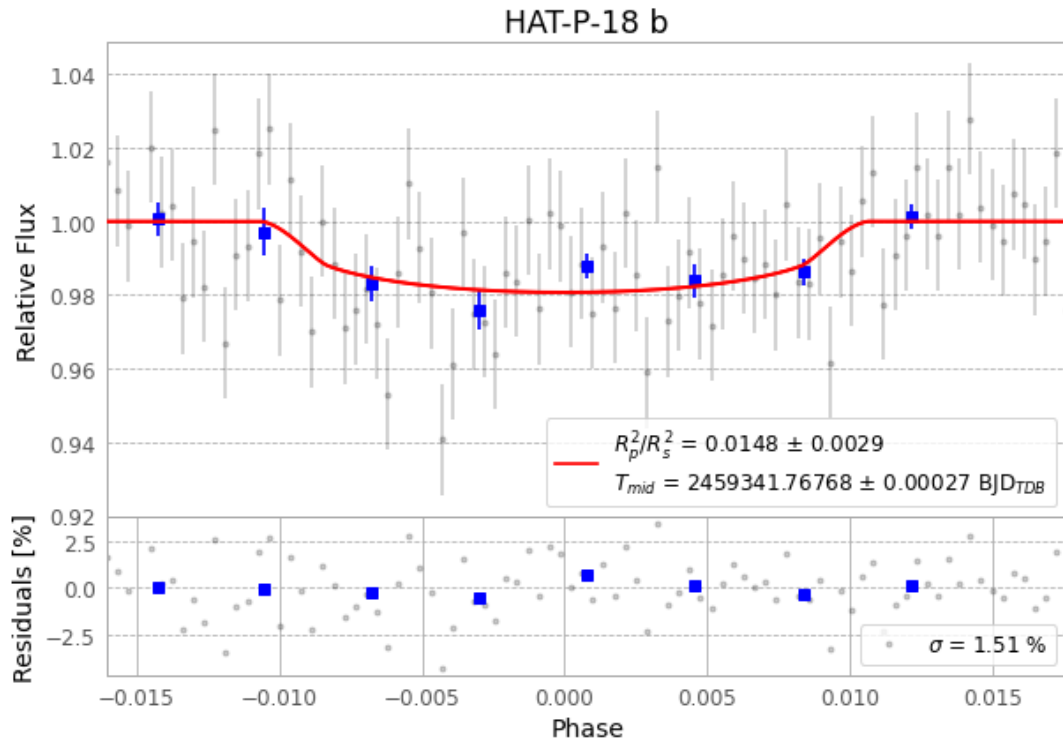


Figure 1 — FinalLightCurve_HAT-P-18 b_2021-05-06.png (SHA-256 18a9e2554bd36bd0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [276, 252], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 fbc6873d4468bde5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

90 FITS frames (59 MB), HATP-18210507041212.FITS → HATP-18210507083912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-18-210507.log (59c83d2cf1ec...), FinalLightCurve_HAT-P-18 b_2021-05-06.png (18a9e2554bd3...), FinalLightCurve_HAT-P-18 b_2021-05-06.csv (4758fa5fd408...)

Compute resources

Wall time 138.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-15 00:59Z.